

Model Analysis Report

Flood Classifier Model

MKM-FC-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Classifier sensitivity to n_estimators (gauge GAUGE-082c, max_depth=4, lr=0.1).

n_estimators	AUC-ROC	Accuracy	Production
50	0.8010	0.7936	
100	0.8002	0.7917	
200	0.7965	0.7865	
500	0.7931	0.7824	

Table 2: Classifier sensitivity to max_depth (gauge GAUGE-082c, n_estimators=200, lr=0.1).

max_depth	AUC-ROC	Accuracy	Production
2	0.7974	0.7904	
3	0.7975	0.7878	
4	0.7965	0.7865	
5	0.7951	0.7840	
6	0.7928	0.7801	

Table 3: Classifier sensitivity to learning_rate (gauge GAUGE-082c, n_estimators=200, max_depth=4).

learning_rate	AUC-ROC	Accuracy	Production
0.0100	0.7950	0.7869	
0.0500	0.8005	0.7901	
0.1000	0.7965	0.7865	
0.2000	0.7902	0.7811	
0.5000	0.7794	0.7763	

Table 4: Classifier sensitivity to subsample fraction (gauge GAUGE-082c, n_estimators=200, max_depth=4, lr=0.1).

subsample	AUC-ROC	Accuracy	Production
0.5000	0.7972	0.7891	
0.6000	0.7976	0.7853	
0.7000	0.7971	0.7897	
0.8000	0.7965	0.7865	
0.9000	0.7985	0.7881	
1.00	0.7975	0.7862	

Table 5: Feature importance at production hyperparameters — v1.1 vs v1.2 (gauge GAUGE-082c).

Feature	v1.1	v1.2
water_level	0.8607	0.6260
hour	0.1393	0.2590
delta_w	—	0.0960
delta2_w	—	0.0190
AUC-ROC	0.7965	0.9720

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 6: Stress testing scenarios for Flood Classifier Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 7: Performance metrics for Flood Classifier Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.